PRODUCT PERFORMACE UPDATE

Application: EVRAZ Rocky Mountain Steel – Porous Plug Extractor

POROUS PLUG EXTRACTOR DETAILS

- **Extractor:** The extractor arm/ram is manufactured out of plain carbon steel pipe housing a solid steel ramming tip. The arm is attached to a pneumatic cylinder allowing the arm to travel in and out of the ladle.

- **Customer Need:** ITC recommended a solution to address two major challenges,
  - Erosion and warping of the steel extractor due to excessive heat.
  - Continual maintenance and replacement.

ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended a hot face coating to address the customers' requirements. The coating design included a proprietary high temperature hot face coating, ITC 213, a proprietary coating specifically formulated for metal surfaces, applied to the steel pipe and ITC 100HT applied as a top coat.

- **High Temperature, Energy Efficient Ceramic Coatings:** ITC manufactures several proprietary, high temperature, energy efficient ceramic coatings to work in concert with refractory or metal substrates. Under ideal conditions, these coatings absorb energy from the process and re-radiate it to the load, imparting the following benefit;

Lower Maintenance & Replacement costs – Protects the steel from oxidation and warping

IMPACT ON THE PROCESS

- EXTRACTOR LONGEVITY
  The current extractor arm replacement rate was twice per week

- REDUCED MAINTENANCE & REPLACEMENT COST

IMPACT ON BOTTOM LINE COSTS

- **EXTRACTOR MAINTENANCE & REPLACEMENT** = $45,000.00 replacing pipe twice a week

- **INCREASED EXTRACTOR LIFE BY 8x**

  Cost to coat extractor arm/ram:

TOTAL IMPACT/YEAR = $